

Title: Electrical and Electronic Component Environmental Requirements

A INDEX

A	INDEX	1
B	SCOPE.....	1
B	SCOPE.....	1
C	OBJECT	2
D	REVIEW	2
E	RELATED DOCUMENTS	2
F	GENERIC REQUIREMENTS.....	3
F 1	Climatic	3
F 2	Lifetime.....	4
F 3	Voltage	5
G	COMPONENT SPECIFIC REQUIREMENTS	7
G 1	Temperatures	8
G 2	Water Ingress	9
G 3	Dust Ingress	9
G 4	Ice Water Shock	10
G 5	Vibration	10
G 6	Mechanical Shock/Drop (Assembly and Shipping)	11
G 7	Mechanical Shock (Closures)	12
G 8	Salt Mist Corrosion	12
G 9	Chemical Resistance	13
H	TESTING REQUIREMENTS	15
H 1	Test Methods.....	15
H 2	Test Sequences & Tests	15
H 3	Test Reports.....	18
I	DOCUMENTATION REQUIREMENTS	19
I 1	Prior to DV	19
I 2	For DV	19
I 3	Prior to PV	20
I 4	For PV	20
I 5	Post PV	21
J	DEFINITIONS.....	22
Z	CHANGE HISTORY	34

B Scope

This standard specifies the minimum Non-EMC Environmental Requirements for electrical and electronic components for Jaguar Land Rover vehicles.

Title: Electrical and Electronic Component Environmental Requirements

C **Object**

This document is designed to assist JLR component areas to specify operational and functional component requirements of their component and to help in the generation of Environmental Test Plans. The requirements in this document are validated using the test methods of TPJLR.18.125.

D **Review**

This document must be reviewed at least every three years from the date of the last review or update.

E **Related Documents**

Table 1 - Related Documents

Ref.	Document	Title	Location
1	TPJLR.18.125	Electrical and Electronic Component Environmental Compatibility Test	Jaguar Land Rover Standards Website
2	STJLR.00.5056	Wet and Dry Zone Definition	Jaguar Land Rover Standards Website
3	STJLR.00.5164	Vehicle Zone Definition	Jaguar Land Rover Standards Website
4	JLR-HTG-559630	Thermal Management Package Guidelines	RMDV ² / SDOTs
5	ISO 20653	Road vehicles - Degrees of protection (IP code) - Protection of electrical equipment against foreign objects, water and access	International Standards Organisation Website
6	ISO 16750	Road vehicles – Environmental conditions and testing for electrical and electronic equipment	International Standards Organisation Website
7	JLR-TMP-561681	High Temperature Endurance Calculator	JLR Way
8	JLR-HTG-564459	EE Environmental Test Selection Matrix	JLR Way

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Title: Electrical and Electronic Component Environmental Requirements

F Generic Requirements

Components shall be designed to survive the environmental extremes encountered in Jaguar Land Rover vehicles.

Environmental requirements for electrical and electronic components and their associated acceptance criteria shall be defined in the Product Design Specification (PDS).

F 1 Climatic

JLR vehicles may be exposed to the following climatic extremes.

a	Relative Humidity range	0 to 100% RH
b	Minimum Air Pressure	53 kPa (equates to 5,500 m above sea level)
c	Maximum Air Pressure	106 kPa (equates to 400 m below sea level)
d	Max Solar Loading	1000 W/m ²
e	Minimum Air Temperature	-40 °C
f	Maximum Air Temperature	55 °C
g	Maximum Shipping Temp	85 °C
h	Maximum Respray Temp	85 °C

Component requirements are a product of vehicle level requirements, based upon the components packaging location. These requirements are detailed on a location-by-location basis. See Section G.

F 1.2 Where the technologies used within the component are sensitive to extremes of either operational or shipping altitude(s), targets shall be agreed between the JLR Component Engineer, Supplier and Vehicle Programme Team.

Title: Electrical and Electronic Component Environmental Requirements

F 2 Lifetime

Components shall be designed to survive in worst case market environmental conditions for a period of 15 years (131,400 hours).

Worst case climatic factors shall be considered for different failure mechanisms, such as worst-case temperature and humidity for CAF failures or worst-case temperature for degradation of Electrolytic Capacitors

In addition:

F 2.1 Components shall be designed to survive a vehicle driven distance of 161,000 kilometers (100,000 miles).

F 2.2 Components that are powered from an ignition or permanent battery feed and **not** powered during EV charging shall be designed to survive a minimum of 8,000 hours in an operational state.

F 2.3 For components that are not powered from an ignition or permanent battery feed, the lifetime operational hours requirement shall be determined by use case analysis and included in the PDS as part of sourcing

F 2.3.1 Where a temperature mission profile is cascaded in the PDS, the L-HTE High Temperature Endurance test parameters may be determined by the completion of JLR-TMP-561681. This shall be completed by the supplier and submitted to Jaguar Land Rover for approval

F 2.4 Components that are operational during EV charging shall be designed to survive 38,000 hours in an operational state

F 2.4.1 The temperature mission profile is to be determined from either vehicle data or vehicle simulation combined with real world climatic data and included in the PDS as part of sourcing.

F 2.4.2 High temperature lifetime test parameters shall be determined by the completion of JLR-TMP-561681. This shall be completed by the supplier and submitted to Jaguar Land Rover for approval.

F 2.5 Components shall be designed to survive a minimum of 12,000 drive cycles

F 2.6 Where Components rated QM according to ISO 26262 cannot meet the above lifetime requirements due to technological limitations alternative targets may be agreed between the supplier, component engineer and the vehicle programme team.

Title: Electrical and Electronic Component Environmental Requirements

F 3 Voltage

- F 3.1 Component Guaranteed Functional voltage range and Guaranteed Performance voltage range shall be selected from Table 2 and added to the component PDS.

Special attention shall be taken to the voltage drop in harnesses.

Table 3 indicates the Voltages that can be expected (in a 12 V system) under different operating conditions.

- F 3.1.1 Where Components are powered from a regulated supply from another module, appropriate voltage targets shall be determined based off the performance of the power source.
- F 3.1.2 Where Components are connected to high voltage feeds such as 48V, 400V or 800V appropriate voltage targets shall be determined based of the performance of these power sources.

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Title: Electrical and Electronic Component Environmental Requirements

Table 2 - Vehicle Supply Voltage Ranges – Functions

Component Test Voltage	Function Performed with Engine Running		Functions Performed with Ignition Off (Battery Supply Only)	
	Requirement for Component in the Vehicle Application	Voltage V _{DC} (Note 1)	Requirement for Component in the Vehicle Application	Voltage V _{DC} (Note 1)
Low Voltage of Guaranteed Function V _{min}	1 Must remain functional in a stalled or a near-stalled engine idle condition.	6	1 Must remain functional when battery is in a low state of charge	9 or 10
	*2. Must remain functional during all engine idle conditions including when vehicle electrical exceeds charging system capacity.	8 or 9	2 Loss of function is acceptable when battery is in a low state of charge or under heavy electrical load.	11
	3 Loss of function is acceptable under condition described in (1) and (2) above.	10		
High Voltage of Guaranteed Function V _{max}	1. Must function during engine warm-up period after cold start (-40 °C to -30 °C).	15.5 or 16	1 Must remain functional when a battery charger is connected to the charging system.	15 or 16
	2 Function not required during engine warm-up period after cold start (-40 °C to -30 °C).	15 or 15.5	2 Loss of function is acceptable when a battery charger is connected to the charging system.	14
Low Voltage of Guaranteed Performance V _{pmax}	1. Must perform to specification in a stalled and/or near-stalled engine condition.	6 or 7	1 Must perform to specification when the battery is in a low state of charge.	10 or 11
	*2. Must perform to specification during all idling conditions (including cases when electrical load exceeds charging system capacity.)	8, 9 or 10	2 Performance to specification is not required when the battery is in a low state of charge.	11
	3. Performance to specification not required under conditions. (1) and (2) above.	10, 11 or 12		
High Voltage of Guaranteed Performance V _{pmin}	1. Must perform to specification during engine and warm-up period after cold start (-40 °C to -30 °C).	15, 15.5 or 16	1. Must perform to specification when a battery charger is connected to the vehicle charging system.	15 or 16
	2. Performance to specification not required during warm-up period after cold start (-40 °C to -30 °C).	14.5 or 15	3. Performance to specification is not required when a battery charger is connected to the vehicle charging system.	13 or 14
Note 1: Select the most appropriate component voltage requirements for Performance and Function from the table while also considering the following items: - Component and subsystem Function Safety Classification and Functional Performance Classification. - Subsystem configuration and component requirements for orderly shut-down at low voltage, fault tolerance, etc. - The charging system performance characteristics for all vehicle applications and operating conditions (also see Table 3). - Voltage drops which may be expected in wiring/distribution systems in all vehicle applications. - Low Voltage for Guaranteed Performance shall be no less than High Voltage for Guaranteed Function. - Low Voltage for Guaranteed Performance shall be no greater than Low Voltage for Guaranteed Function. - Subsystems/components required to operate when the engine is running and not running must meet both requirements; i.e., Voltages must be specified considering both sides of the table appropriately. Note 2: Tolerance on voltage values ±0.2V. *This condition is considered normal and can be expected to occur occasionally in any vehicle.				

Title: Electrical and Electronic Component Environmental Requirements

Table 3 - Vehicle Supply Voltage Ranges - Operating Conditions

Description of Vehicle Conditions				
Ignition-On/Engine Running		System Voltage (VDC) Measured at Battery*	Engine Not Running (Battery supply only)	
Vehicle Operating Condition	Frequency of Occurrence		Vehicle Operating Condition	Frequency of Occurrence
Highest voltage regulation occurs during engine warm-up periods following cold starts at extreme low temperatures.	Common during winter in cold climates.	16.0 15.5	Up to 16 V is possible when a battery charger is connected to the vehicle charging system.	Common service practice to recharge battery.
Normal driving and idling with typical electrical system loading.	Often.	15.0 14.5 14.0		
		13.5 13.0	Normal voltage range for a battery with a normal state of charge and typical electrical loading.	Often.
System voltage drops out of regulation when electrical load exceeds charging system capacity	Common when idling in gear with accessories in use.	12.5 12.0	Battery is in a low state of charge or is under heavy electrical loading. (Will still start engine).	Occasionally.
		11.5 11.0 10.5	Battery is in a low state of charge and may have dead cells. (Will not start engine if below approx. 10.5 V).	Occasionally.
	Rare. Extreme loading.	10.0 9.5 9.0 8.5		
	In engine stall or near-stall conditions system voltage may dip below 9 volts and then return to a steady state condition.	Very Rare.	8.0† 7.5 7.0 6.5 6.0	
†Engines are usually not guaranteed to run below 8 V.				
*Voltage at the component will be equal to or lower than the voltage at the battery depending on the amount of voltage drop within the wiring distribution system. Voltage ranges given are approximate (±0.5 V to ±0.75 V). Actual voltage will depend on individual vehicle design and will differ with normal production variations in charging system component characteristics.				

G Component Specific Requirements

Vehicle Components shall be designed to survive the environmental extremes encountered in their packaging location(s).

Environmental requirements for electrical and electronic components and their associated acceptance criteria shall be defined in the Product Design Specification (PDS).

If the component is to be used on multiple vehicles, or multiple packaging locations, then the location(s) in each vehicle shall be determined. The widest operational temperatures and voltages shall then be chosen and the harshest applicable tests shall be selected for the Environmental Test Plan.

Title: Electrical and Electronic Component Environmental Requirements

G 1 Temperatures

The Component temperature requirements listed in Table 4 shall be defined in the PDS.

G 1.1 Minimum temperatures T_{min} , and T_{Emin} shall be defined as -40°C .

G 1.2 Maximum temperatures T_{max} and T_{Emax} shall be defined by consulting the Thermal Management Package Guidelines (JLR-HTG-559630).

G 1.2.1 Where the temperature requirements in the Thermal Management Package Guidelines cannot be met for due to technology limitations, alternative requirements shall be agreed with the TASE PAT leader or Thermal Technical Specialist based on either measured vehicle data or vehicle thermal simulations.

G 1.2.2 The maximum exposure temperature T_{Emax} shall be no less than 85°C to account for maximum temperatures during shipping and paint respray stoving temperature for exterior parts.

G 1.2.3 The maximum exposure temperature T_{Emax} shall be no less than the maximum operating temperature T_{Max} .

G 1.2.4 Where the maximum temperature of a component is determined by the performance of an active cooling system that allows the component's temperature to be maintained below that of the packaging environment, the T_{max} may be determined based of the performance of the cooling system. The value of T_{Emax} and T_{Soak} shall be determined from the package location.

G 1.3 The Performance Operating Temperature Range T_{Pmin} to T_{Pmax} shall be the same as the Operating Temperature Range (T_{min} to T_{max}) unless there are technological reasons for the Performance Operating Temperature Range to be limited.

In this case restricted values for T_{Pmin} and T_{Pmax} shall be defined after consultation between the Component Owner, Vehicle Programme team and Supplier with TASE input (where required). This Performance Operating Range shall be stated in the PDS with the rationale.

Title: Electrical and Electronic Component Environmental Requirements

Table 4 - Temperature Requirements Definitions

Temp	Definition
T_{min}	Minimum Operating Temperature The minimum temperature at which the component shall operate.
T_{max}	Maximum Operating Temperature The maximum temperature at which the component shall operate.
T_{Pmin}	Minimum Performance Operating Temperature The minimum temperature that the component meets its full functional performance specification.
T_{Pmax}	Maximum Performance Operating Temperature The maximum temperature that the component meets its full functional performance specification.
T_{Emin}	Minimum Exposure Temperature The minimum temperature the component shall survive in a non-operational state. Includes assembly, shipping and non-operational periods in the vehicle.
T_{Emax}	Maximum Exposure Temperature The maximum temperature the component shall survive in a non-operational state. Includes assembly, shipping and non-operational periods in the vehicle. For exterior components it also includes paint repair temperatures.
T_{soak}	Maximum Soak Temperature The maximum temperature experienced after a powertrain or solar soak.

G 2 Water Ingress

The Component IP rating(s) shall be defined in the PDS in line with the requirements of STJLR.00.5056.

G 2.1 Where components have external moving parts, the PDS shall define whether these shall be in motion during testing.

A summary of this IP Rating(s) is shown in [Appendix C](#).

G 3 Dust Ingress

The Component IP rating(s) shall be defined in the PDS in line with the requirements of STJLR.00.5056.

A summary of this IP Rating(s) is shown in [Appendix C](#).

Title: Electrical and Electronic Component Environmental Requirements

G 4 Ice Water Shock

Components packaged in the following zones according to STJLR.00.5164

- B (Engine Bay),
- D (Vehicle Exterior)
- E (Wheels and Suspension)

Shall be designed to withstand rapid changes in temperature by the action of being splashed by, or immersed in, cold water.

This shall be validated by performing the Ice Water Splash Test or Ice Water Immersion Test. Test selection shall be determined from Table 5 using the specified IP rating.

Table 5 - Applicability of Ice Water Tests

Packaging Zone	IPX3 – IPX6K or IPX9K	IPX7-IPX8
B (Engine Bay) D (Vehicle Exterior) E (Wheels and Suspension)	Complete Ice Water Splash Test	Complete Ice Water Immersion Test
A (Passenger Compartment) C (Load space)	-	-

Package locations are summarized in [Appendix B](#).

G 5 Vibration

The Component shall be designed to meet the requirements of vibration testing. The required profile(s) from TPJLR.18.125 shall be cascaded in the PDS.

- a Body Mounted (Sprung Mass)
- b Suspension/Axle Mounted (Unsprung Mass)
- c Engine Mounted
- d Gearbox Mounted
- e Flexible Plenum Chamber

G 5.2 Vibration profiles for components mounted on vibration inducing parts not covered by the above profiles (such as Pumps, Motors or EDUs, Differential Units or parts mounted on Harnesses or Hoses) shall be agreed between the JLR Component Engineer, D&R PAT Leader/Technical Specialist and the Supplier based upon measured or simulated data. These profiles shall be included in the PDS.

G 5.3 For body mounted components that are not mounted directly to the Body in White, it shall be ensured that the bracket/mounting strategy does not expose the component to levels of vibration that exceed the TPJLR.18.125 Body Mounted profile when the system is excited with the TPJLR.00.047 Body Mounted profile.

G 5.3.1 This above shall be verified using appropriate CAE software tools and models, and validated by measurement of the vibration level at the component interface during system level tests.

Title: Electrical and Electronic Component Environmental Requirements

G 6 Mechanical Shock/Drop (Assembly and Shipping)

The Component shall be designed to survive an agreed level of shock / drop events that can occur during production and shipping. The appropriate shock / drop tests shall be selected from Table 6 and defined in the PDS

G 6.1 Applicable Process FMEAs shall include parts being dropped during transportation and assembly with appropriate measures taken in line with the level of Shock/Drop robustness and testing completed.

Table 6 - Mechanical Shock/Drop Test Applicability

		Mechanical Shock/Drop Test Requirements See Note 2					
Description	Example Component Types	Package Drop	Handling Drop	Medium Mech. Shock	Low Mech. Shock	Required Component Labelling	Processing/ Handling Approach
Non-robust Component Design/ Careful Handling & Protective Packaging	Instrument clusters, displays.	✓ (Note 1)	N/A	N/A	✓	"Set aside If Dropped"	Minimize rough handling. Control damaged material.
Semi-robust Component Design Large Mass	Electronic modules, large mass.	✓ (Note 1)	N/A	✓	N/A	"Set aside if Dropped"	Minimize rough handling. Control damaged material.
Robust Component Design/ Minimal Handling Precautions	Encased sensors, small modules.	N/A	✓	N/A	N/A	None	No special controls.
Note 1: Packages designed for human handling (18 kg max) shall be subject to Package Drop testing. Note 2: Package Drop and Handling Drop requires a sample for each corner and side, for most components this will be 14 test samples, (6 sides and 8 corners)							

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Title: Electrical and Electronic Component Environmental Requirements

G 7 Mechanical Shock (Closures)

The component shall be designed to survive a lifetime of shock events. The number of shocks, the acceleration and duration shall be defined in the PDS.

The shocks applied shall be in the form half sine impulses, with an amplitude of 491 m/s² and a duration of 11 ms

The minimum number of shocks used in the Bump Test relative to the Component vehicle location is defined in Table 7. A larger number of shocks than indicated may be selected as agreed between the Component Owner and the Supplier.

Table 7 - Mechanical Shock Requirements

Location (on or immediately surrounding)	Bump Test Number of Shocks
Driver's door, cargo door	13,000
Passenger doors	6,000
Boot lid, tailgate	2,400
Bonnet	720

G 8 Salt Mist Corrosion

The component shall be designed to meet the requirements of Salt Mist Testing which represents coastal atmospheric and road salt effects. Salt Mist Atmosphere test durations are dependent on the Component Packaging Zone as described in STJLR.00.5164 and shall be selected from Table 8 and documented in the PDS.

Table 8 - Salt Mist Atmosphere Test Durations

Component Packaging Zone	Classification	Test Duration
A13a, A13b and A14 (NOTE)	II	24 h
C4a and C4b	II	24 h
D11 and D12	II	24 h
D1 to D3	III	48 h
B1 to B6	IV	168 h
D4 to D10 and D13	IV	168 h
E1 to E4	IV	168 h
All Other Locations	I	Test N/A

NOTE: Where the component is completely packaged behind trim in interior zones this test may be waived on agreement with the JLR Component Engineer. Written agreement shall be sought. This does not include components where air is drawn from the cabin, either actively or through convection.



Title: Electrical and Electronic Component Environmental Requirements

G 9 Chemical Resistance

The component shall be resistant to any chemicals that it may reasonably be exposed to during its operational life.

These chemicals along with their application methods shall be determined from Table 9 and documented in the PDS.

G 9.1 Additional chemicals may be tested as agreed between the supplier and the JLR Component Engineer.

The applicable chemicals and relevant component surfaces to be tested should be based on analyses of the local market products and automotive fluids specified by the programme.

G 9.2 For other packaging locations, the chemical reagents to be tested, and the application methods shall be defined by the JLR component engineer.

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Title: Electrical and Electronic Component Environmental Requirements

Table 9 - Chemicals for Test

Chemical Reagent (Note 3)	Vehicle compartments/locations (Package Zone)				Application Methods (Note 1)	Test Temp. (Note 2)	Test Duration
	Engine (B*)	Exterior (D*), (E*)	Luggage (C*)	Passenger (A*)			
Diesel Fuel	x				I,III,IV,V	T _{Emax}	22:00
"Bio" Diesel	x				I,III,IV,V	T _{Emax}	22:00
Unleaded Petrol/Gasoline	x				I,III,IV,V	T _R	00:10
Methanol	x				II,III,IV,V,VI	T _R	00:10
Engine Oil	x				II,III,IV,V	T _{Emax}	22:00
Differential Oil	x				II,III,IV,V,VI	T _{Emax}	22:00
Transmission Fluid	x				II,III,IV,V,VI	T _{Emax}	22:00
Hydraulic Fluid	x				II,III,IV,V	T _{Emax}	22:00
Greases	x				II,III	T _{Emax}	22:00
Silicone Oil	x				I,II,III,V	T _{Emax}	22:00
Cleaning Solvent	x				I,II,III	T _R	00:10
Contact Spray	x				I,II,III	T _R	22:00
Antifreeze Fluid	x				I,III,IV,V,VI	T _{Emax}	22:00
Calcium Chloride (Solution)	x	x			I, II, III	T _R	22:00
Urea	x	x			II,III,V	T _{Emax}	22:00
Protective Lacquer	x	x			I,II	T _R	22:00
Protective Lacquer Remover	x	x			I,III,IV,V	T _{Emax}	22:00
Runway De-icer	x	x			I,II,IV	T _R	02:00
Vehicle Washing Chemicals	x	x			I,II,III,IV,V	T _R	02:00
Window Screen Washer Fluid	x	x	x		II,III,IV,V	T _R	02:00
Cold Cleaning Agent	x	x	x		I,II,III,IV,V,VI	T _R	22:00
Brake Fluid	x		x		II,III,V	T _{Emax}	22:00
Battery Fluid (Note 4)	x			x	III,V	T _R	22:00
Denatured Alcohol	x	x	x	x	I,II,III,IV,V	T _R	00:10
Cavity Protection		x			II,III	T _R	22:00
Damper Oil		x			I,II,III,IV,V	T _R	22:00
Wheel Cleaner		x			I,II,III,IV	T _R	02:00
Glass Cleaner		x	x	x	I,III	T _R	02:00
Ammonia Containing Cleaner		x	x	x	II,III,V	T _R	22:00
Leather Wax		x	x	x	II, III	T _R	22:00
Kerosene			x		II,III,IV,V	T _R	00:10
Deodorizer Spray			x	x	I	T _R	22:00
Interior Cleaners			x	x	I,III	T _R	02:00
Acetone				x	I,II,III	T _R	00:10
Perspiration				x	II,III,V	T _R	22:00
Cosmetic Products eg Creams/Moisturisers				x	II,III	T _R	22:00
Refreshment containing Sugar & Caffeine				x	III,IV	T _R	22:00
Cream, Coffee Whitener				x	III,IV	T _R	22:00

Note 1: Application methods: I – Spray, II – Brush, III – Wipe, IV – Pour, V – Dip, VI – Immerse
Note 2: T_R is defined as 23 °C ± 5 °C
Note 3: The listed chemicals and any additional chemicals shall be considered and included in the tests if an exposure risk is identified in the FMEA. Additional chemicals shall be considered by the JLR Component Owner and included in the Product Design Specification.
Note 4: Within a radius of 200 mm from the centre of the battery and below the top of the battery.
Note 5: Definitions of most of the chemicals can be found in ISO 16750-5 : 2011

Title: Electrical and Electronic Component Environmental Requirements

H Testing Requirements

H 1 Test Methods

The requirements in this specification shall be validated as described by the tests in TPJLR18.125.

The tests shall be conducted at locations having the necessary equipment and facilities with accreditation to ISO 17025.

H 1.1 The acceptance of alternative test methodologies, or test data may be agreed between the Component Owner and the Electrical Environmental Technical Specialist where the tests are deemed equivalent or harsher. Written acceptance shall be sought. For example COTS components tested to ISO 16750-3/4/5.

H 1.2 A test report shall be generated for each completed test (pass or fail).

H 1.3 In the event a DUT fails to meet a requirement as part of the acceptance criteria for a test the JLR Component Owner shall be notified within 72 hours of the discovery.

H 2 Test Sequences & Tests

The test sequences and individual tests for Environmental Validation shall be selected from the following sections based upon the different levels of change. Where the Component has been modified refer to section H 2.3.

NOTE: Performance Evaluation (L 1), Functional Evaluation (L 4) and Internal Inspection (L 6) refer to Acceptance Tests in TPJLR.18.125.

Tests are grouped into

- **CL – Climatic Tests**
- **ME – Mechanical Tests**
- **L – Lifetime Tests**
- **IN – Ingress Tests**

Test sequences and individual tests shall be preceded by:

Performance Evaluation (L 1) + Functional Evaluation (L 4)
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and shall be followed by:

Performance Evaluation (L 1) + Functional Evaluation (L 4)
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Internal Inspection (L 6)

Title: Electrical and Electronic Component Environmental Requirements

H 2.1 New Design/Major Modification (Full Environmental) Tests

The following sequences are used for the Full Environmental set of tests along with individual tests listed below. Sequences 1 and 2 can be combined into a single sequence, as indicated, if availability of parts is limited.

Sequence 3 contains preconditioning tests for L-CD Controls Durability and L-MW Mechanical Wear-out. Where components have neither controls nor moving mechanical elements, this sequence may be waived.

Sequence 1 Tests - 12 samples
CL-LTE - Low Temperature Exposure *
CL-HTE - High Temperature Exposure *
CL-LTO - Low Temperature Operation *
CL-HTO - High Temperature Operation *
CL-TST - Temperature Step Test *
CL-PTC - Powered Thermal Cycle *
CL-TSA - Thermal Shock in Air
Performance Evaluation (L 1) + Functional Evaluation (L 4)
ME-V - Powered Vibration
ME-S - Mechanical Shock/Drop (14 samples required)
AN - Audible Noise
CL-THC - Temperature Humidity Cycle
IN-TSI - Ice Water Immersion
IN-TSS - Ice Water Splash
IN-W - Water/Fluid Ingress
IN-D - Dust Ingress
CL-SALT - Salt Mist Atmosphere
<i>If using same parts continue with Sequence 2 tests</i>
<i>* These tests can be performed in any order</i>

Sequence 2 Tests - 12 samples
CR - Chemical Resistance (may require extra samples)
ME-C - Connector Lead/Lock Strength

Sequence 3 Tests - 6 samples
CL-TSA - Thermal Shock in Air
ME-V - Powered Vibration
IN-D - Dust Ingress
L-CD - Controls Durability
L-MW - Mechanical Wear-out
CL-SALT - Salt Mist Atmosphere

Title: Electrical and Electronic Component Environmental Requirements

Individual Tests
CL-CG - Corrosive Gases (6 samples)
L-TSE - Thermal Shock Endurance (8 samples)
CL-DEW - Dewing Test (6 samples)
L-HTE - High Temperature Endurance (6 samples)
L-THB - 85/85 High Temperature – High Humidity Endurance (6 samples)

H 2.2 New Design/Major Modification (Motors/Solenoids without a PCBA)

The following sequence is used for the reduced Full Environmental set of tests along with individual tests listed below for **Motors or Solenoids that do not have a PCBA**.

Sequence 4 Tests - 6 samples
CL-LTE - Low Temperature Exposure *
CL-HTE - High Temperature Exposure *
CL-LTO - Low Temperature Operation *
CL-HTO - High Temperature Operation *
CL-TSA - Thermal Shock in Air
Performance Evaluation (L 1) + Functional Evaluation (L 4)
ME-V - Powered Vibration
ME-S - Mechanical Shock/Drop (14 samples required)
AN - Audible Noise
IN-TSI - Ice Water Immersion
IN-TSS - Ice Water Splash
IN-W - Water/Fluid Ingress
CL-SALT - Salt Mist Atmosphere
<i>* These tests can be performed in any order</i>

Sequence 5 Tests – 6 Samples
CL-TSA - Thermal Shock in Air
ME-V - Powered Vibration
ME-S - Mechanical Shock/Drop (14 samples required)
IN-D - Dust Ingress
L-MW - Mechanical Wear-out
CL-SALT - Salt Mist Atmosphere

Individual Tests
ME-C - Connector Lead/Lock Strength
CR – Chemical Resistance

Title: Electrical and Electronic Component Environmental Requirements

H 2.3 Other Modifications

Where a part is being revalidated because of a modification, the appropriate test sequence shall be selected from JLR-HTG-564459.

Where the changes are not wholly described by one or more of the sections in JLR-HTG-564459 a custom test sequence shall be agreed between the Supplier, Component Owner, and Environmental Technical Specialist

H 3 Test Reports

Test reports shall be provided in English and include the following:

- Complete test procedure - including sequence of steps, stimuli and test software used.
- Details of the test equipment used (e.g. power supplies, environmental chambers and instrumentation) including:
 - manufacturer
 - model
 - serial number
 - last calibration date
 - next calibration date.
- Design of load boxes and test boxes (circuits and BOMs) used with photographs.
- Photographs and diagrams of the test set up to include sufficient information to exactly re-construct the test at a later date.
- Measurements of the test laboratory conditions (i.e. temperature, humidity).
- Measurements of test conditions (e.g voltage, temperature, current)
- Records of test results including oscilloscope traces.
- Photographs of parts before and after testing, clearly highlighting any test failures or visual changes.
- Identification of non-compliances

Title: Electrical and Electronic Component Environmental Requirements

I Documentation Requirements

The supplier shall deliver the documentation as described in this requirement to the JLR Component Owner for each specified stage of the development of the component.

I 1 Prior to DV

- Signed off WCCA.
- Design Verification Plan (DVP) to include test sample sizes and timing plan.
- Definition and agreement on the functional tests to be performed during TPJLR.18.125 acceptance tests Performance Evaluation L 1, Functional Check L 2, Monitored Operation L 3 and Functional Evaluation L 4 with measurements to be taken, loads to be used and acceptance criteria.
- Evidence that components used in the DUTs meet the 1,000 hour 85 %RH/85 °C requirement.
- Confirmation that components used in the DUTs meet the maximum and minimum temperature requirement of the unit (Ambient and Internal when specified).
- Reflow, Wave, and/or Selective Solder temperature profiles with written verification that the profiles meet the solder and component manufacturers' thermal requirements.
- Planned End of Line (EoL) test measurements and how they correlate to the JLR Product Design Specification (PDS).
- HALT test results (where HALT testing is being performed) with 8Ds of any failures/discrepancies.
- Where parts will be reworked sample sizes shall be doubled with the additional samples being representative of reworked parts.
- Cleanliness measurement data on incoming bare boards and PCBAs using the IPC-TM-650 2.3.28 Ion Chromatography or IPC-TM-650 2.3.25 ROSE test method. Results are to be reported in µg/cm² by element content or NaCl equivalent.
- PCBA X-ray inspection/PCBA sectioning of solder joints including 8Ds on non-compliances to applicable sections of IPC-A-610 "Acceptable Class 3" or better must be met.

I 2 For DV

- Completed DVP&R with test data summary and pictures of solder sections/inspections.
- Copy of the test log signed by the supplier test and design engineers. Where parameters are required to be measured as part of the test requirements, both at set points or continuously logged, these shall be included in the test logs.
- Full test reports in English detailing:
 - The test facility and the individuals performing the test.
 - Diagrams and photographs of the test setup (sufficient to replicate at a future date).
 - A list of the test equipment used including type, serial number and calibration data.
 - The test procedure used.

Title: Electrical and Electronic Component Environmental Requirements

- The report shall be signed (either physically or digitally) by the Engineer performing the test and the test laboratory manager.
- The test results.
 - An 8D for each failure/discrepancy.
 - Rolled Throughput Yield for DV run.
 - Results of a study of the micro-strain the board experiences during the prototype build process and a comparison to those acceptable to the parts. The strain measured must be less than 50 % of that acceptable to the parts.
 - If boards are built using the production process provide Cleanliness Measurement data for the PCBA using the IPC-TM-650 2.3.28 Ion Chromatography or IPC-TM-650 2.3.25 ROSE test method. Results are to be reported in $\mu\text{g}/\text{cm}^2$ by element content or NaCl equivalent.

I 3 Prior to PV

- Process Verification Plan (PVP) with timings and sample sizes.
- Where parts will be reworked sample sizes shall be doubled with the additional samples being representative of reworked parts.
- Reflow, Wave, and/or Selective Solder temperature profiles with written verification that the profiles meet the solder and component manufacturers' thermal requirements.
- Final End of Line test measurements and how they correlate to the JLR Product Design Specification (PDS). A list of requirements not checked at end of line with the reasoning as to their non-inclusion.
- The EOL tester fail safe strategy.
- Copy of the Process Control Plan(s) with a list of planned Audits.

I 4 For PV

- Completed PVP&R with data summary and pictures of solder sections/inspections.
- Copy of the test logs signed by the supplier test and design engineers. Where parameters are required to be measured as part of the test requirements, both at set points or continuously logged, these shall be included in the test logs.
- An 8D for each failure/discrepancy.
- Audit results with the P_{pk} s for each significant characteristic.
- HALT test results (where HALT testing is being performed) with an 8D for each failure/discrepancy.
- List of recommendations for In-Process (IP) testing based upon PV results.
- Results of tester validation and fail safe verification.
- Rolled Throughput Yield for PSW run. For less than 99 % yield a plan shall accompany the data with steps to improve the yield to 99 %.
- Results of a study of the micro-strain the board experiences during the production build process and a comparison to those acceptable to the parts. The strain measured must be less than 50 % of that acceptable to the parts.
- Provide Cleanliness Measurement data for the PCBA. Suggested methods are those recommended by the IPC or equivalent. Results are to be reported in $\mu\text{g}/\text{cm}^2$ by element content.

Title: Electrical and Electronic Component Environmental Requirements

I 5 Post PV

For the first year of production, once every 30 days the current Rolled Throughput Yield shall be sent to the JLR STA engineers as well as the status of the plan to improve the yield to 99 % if not already being achieved.

For the first year of production, once per quarter run a HALT Operation Limit test to the PV operation limits and supply results to the JLR D&R and Reliability engineers along with 8Ds for any failures/non-compliances.

For as long as the DUT is in production the supplier will monitor the Warranty and TGW performance and request part returns from the field through the JLR Component Owner. Returns shall be root-cause analyzed and 8Ds provided.

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Title: Electrical and Electronic Component Environmental Requirements

J Definitions

- J 1.1 **Closure.** Opening into the Body in White (BIW). Normal examples are the boot/trunk, doors, and bonnet.
- J 1.2 **Component.** The electrical/electronic part being supplied to JLR.
- J 1.3 **Component Engineer (CE).** The engineer responsible for the component to be tested.
- J 1.4 **Computer Aided Engineering (CAE).** A broad term that covers the use software to aid in engineering analysis. Includes fields such as Finite Element Analysis (FEA) and Computational Fluid Dynamics (CFD)
- J 1.5 **Commercial Off The Shelf (COTS).** A part that has already been designed by the supplier and intended for multiple automotive OEMs.
- J 1.6 **Design Verification (DV).** The act of determining the vehicle environment (temperature, vibration, duty cycles, loads, etc.) and defining and conducting all tests and evaluations necessary to assure the adequacy of a design for the vehicle environment.
- J 1.7 **Design Verification Plan (DVP).** A test plan which lists all tests and evaluations (including references to the applicable test specifications), acceptance criteria, timing and sample size requirements for Design Verification of a specific product design or design level. Test dates and durations may also be included.
- J 1.8 **Design Verification Plan & Report (DVP&R).** A document based upon the DVP but also summarising the completion status and dates of all tests, and whether each test has been passed or has failed.
- J 1.9 **Design Verification Report (DVR).** A report documenting Design Verification test results and actual test completion dates.
- J 1.10 **Design Verification (DV) Test.** Any test specified in a Design Verification Plan (DVP) or Design Verification Plan and Report (DVP&R) that is intended to verify a design attribute.
- J 1.11 **Development Testing.** Informal testing performed for developmental purposes during the development stages of a program, typically to gain insight into product design attributes such as operating limits, robustness to environmental conditions, etc. Examples of development testing include step-stress testing, characterization testing etc. Also see Pre-DV testing.
- J 1.12 **Device Under Test (DUT).** A DUT is a module/component to be tested in one or more of the tests in this document.
- J 1.13 **Electric Vehicle (EV).** A vehicle which uses electrical power for propulsion. This category includes Plug in Hybrid Electric Vehicles (PHEVs) and Battery Electric Vehicles (BEVs)
- J 1.14 **End of Line (EoL) Tests.** A set of tests, baseShouldn't duty/load cycled upon the Product Design Specification, completed on production parts at the end of the production line.

Title: Electrical and Electronic Component Environmental Requirements

- J 1.15 **Engineering Specification (ES).** A formally released document which, along with associated engineering drawings, operationally defines product design requirements in terms appropriate for use by the manufacturing facility. The Engineering Specification contains all Production Validation (PV), In-Process (IP) and Functional Tests, acceptance criteria and any other items not appropriate or practical to include in an engineering drawing. Also known as the **Product Design Specification (PDS)**
- J 1.16 **Environmental Stress Screen (ESS).** A test, usually performed on 100 % of production samples, which utilizes environmental stresses to stimulate relevant failures for the purpose of screening out substandard (weak) parts, assemblies or sub-assemblies.
- J 1.17 **Finite Element Analysis (FEA).** A mathematical technique for analysing stress where a physical structure is split into substructures called “Finite Elements”
- J 1.18 **Highly Accelerated Life Test (HALT).** An evaluation method to determine the operational and destruct limits of an electrical/electronic design/processed component using special chambers. The environments include but are not limited to min/max temperature, vibration and thermal shock
- J 1.19 **Highly Accelerated Stress Screen (HASS).** A process monitor or audit method that uses the results of the HALT tests to create a screen which can be used to verify minimum operational limits for the environments listed above.
- J 1.20 **In Process (IP) Test.** A test specified in a Product Design Specification (PDS), Product Verification Specification (PV Spec) and/or Control Plan which is intended for use in an IP testing activity.
- J 1.21 **In Process (IP) Testing.** The act of assuring that production parts continue to meet design intent through periodic tests and evaluations.
- J 1.22 **PCBA.** Printed Circuit Board assembly.
- J 1.23 **PCBA Component.** A discrete electrical/electronic part on the PCBA of a Component, such as a Resistor, Capacitor, Diode, FET or Microcontroller.
- J 1.24 **Product Design Specification (PDS).** A formally released document which, along with associated engineering drawings, operationally defines product design requirements in terms appropriate for use by the manufacturing facility. The Product Design Specification contains all Production Validation (PV), In-Process (IP) and Functional Tests, acceptance criteria and any other items not appropriate or practical to include in an engineering drawing.
- J 1.25 **Programme Attribute Team Leader (PAT Leader).** The JLR engineer responsible for creation and sign off of attribute targets for a vehicle programme for a specific attribute team.
- J 1.26 **Rolled Throughput Yield** The probability that a process with more than one step will produce a defect free unit. It is the product of **yields** for each process step of the entire process.
- J 1.27 **Thermal Aerodynamics and Systems Engineering (TASE).** A Vehicle Engineering Department (Attribute Team) in Jaguar Land Rover.
- J 1.28 **Things Gone Wrong (TGW)**

Title: Electrical and Electronic Component Environmental Requirements

J 1.29 **Worst Case Circuit Analysis (WCCA).** (Extreme Value Analysis) is a rigorous mathematical evaluation of circuit performance attributes against performance tolerance limits under simultaneous existence of all the most unfavourable conditions being at realizable limits.

WCCA shall include the following:

- Circuit description
- System schematic
- Bill of Materials
- Theory of operation
- Description of functionality provided by each component
- Specifications
- Environmental conditions (temp, voltage, EMC, ground shifts, interface)
- Component specifications and assumptions
- Analysis schematic (with and without parasitics)
- Worst case analysis
- Stress analysis
- A/C analysis (if required)
- Transient analysis
- Summary
- Bench test results

Title: Electrical and Electronic Component Environmental Requirements

APPENDIX B – SUMMARY OF PACKAGING LOCATIONS

The vehicle location(s) shall be identified from Figures 1 to 13 and Table 4 below and added to the PDS.

The following section is an extract from STJLR.00.5164.

Figure 1 - Interior Zones for Saloon vehicle (up to 1.5 m high)

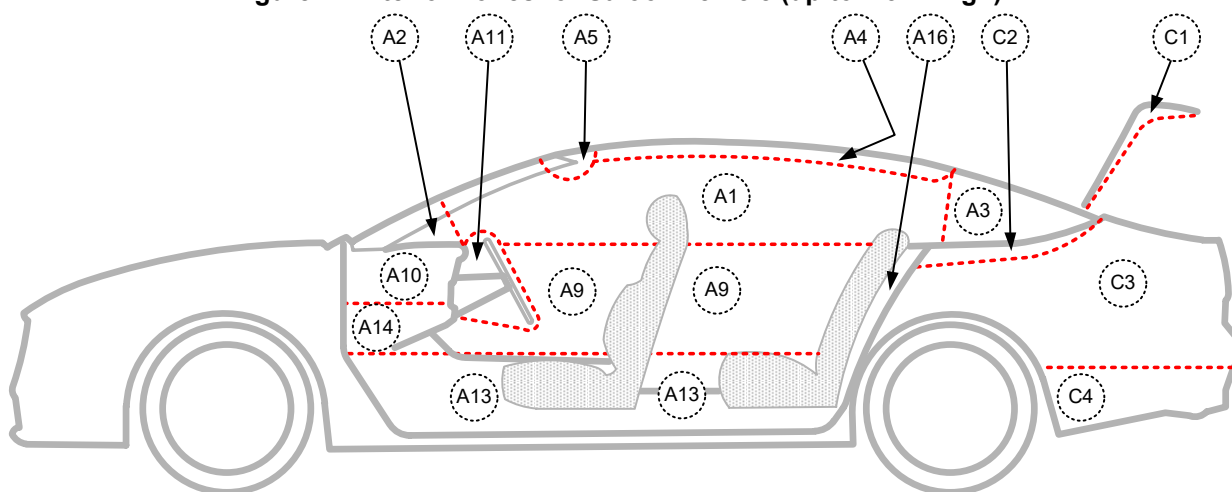
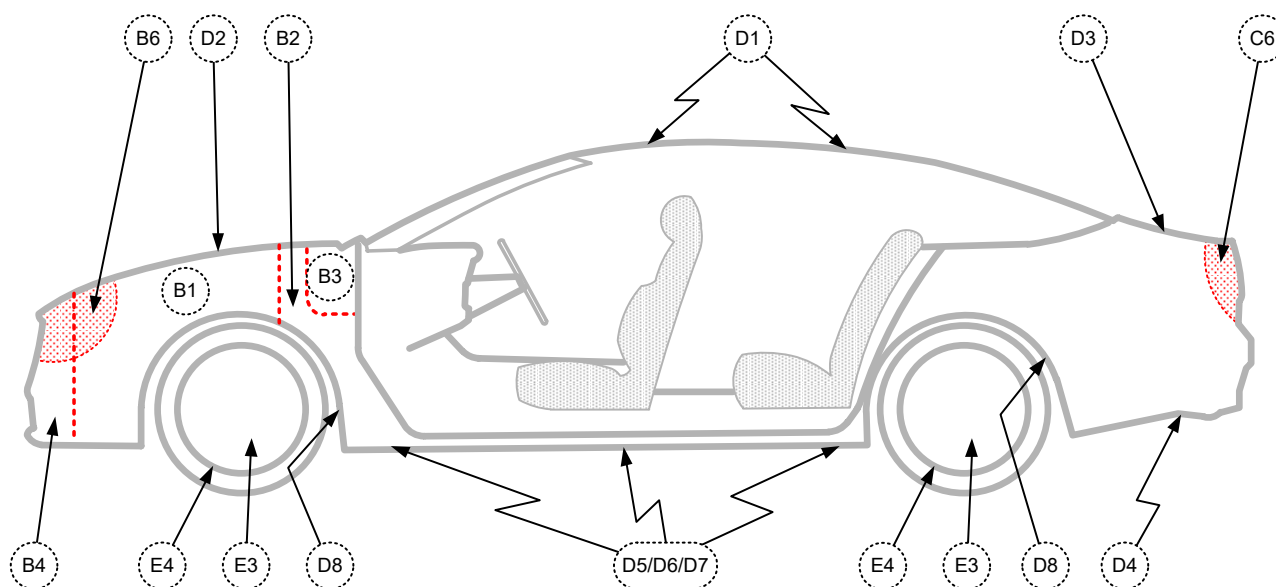


Figure 2 - Exterior zones for Saloon vehicle (up to 1.5 m high)



Title: Electrical and Electronic Component Environmental Requirements

Figure 3 - Interior Zones for Convertible vehicle

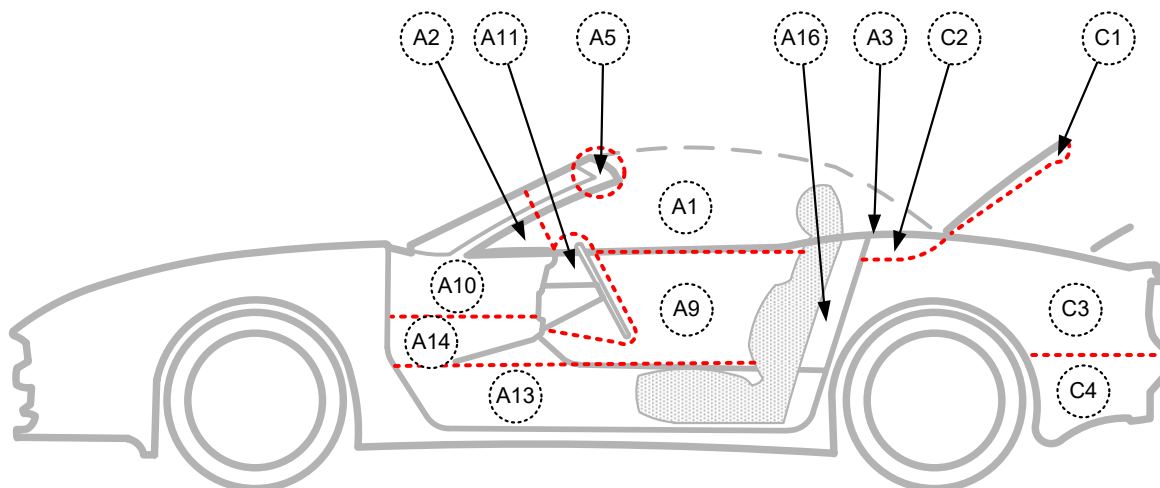
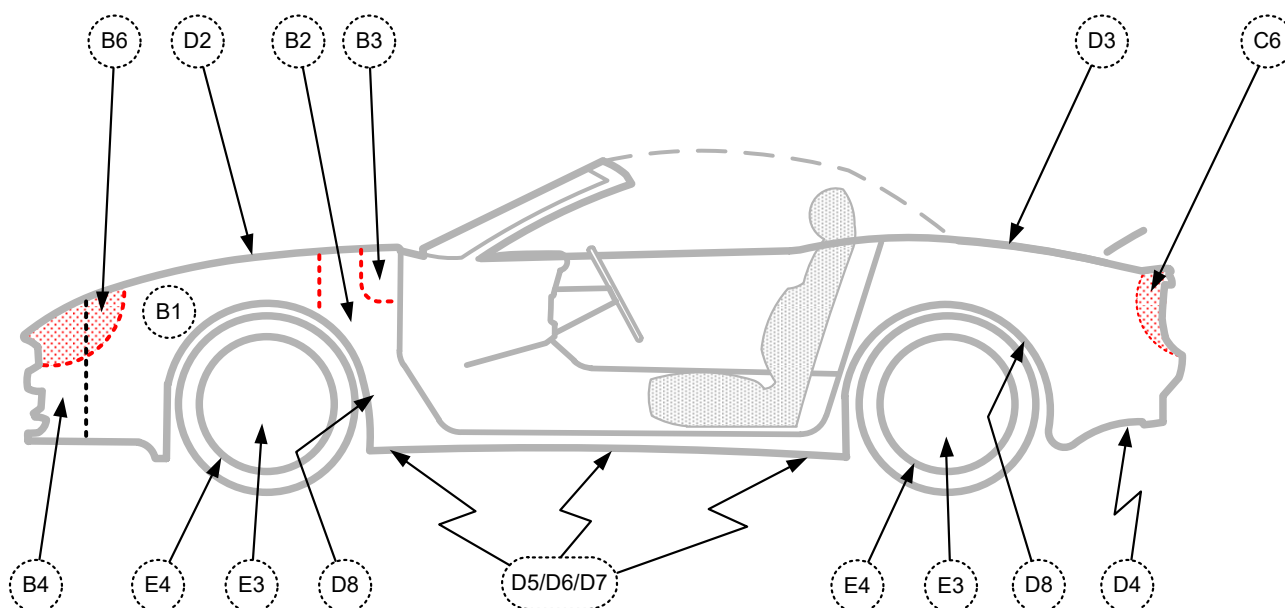


Figure 4 - Exterior Zones for Convertible vehicle



Title: Electrical and Electronic Component Environmental Requirements

Figure 5 - Interior Zones for SUV

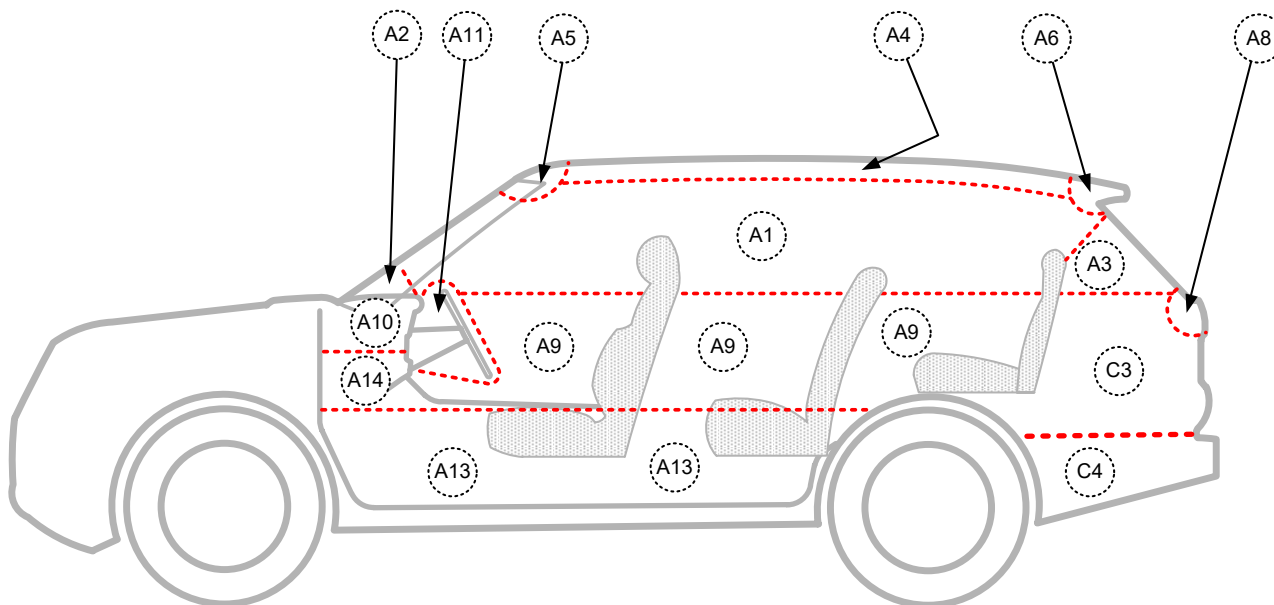
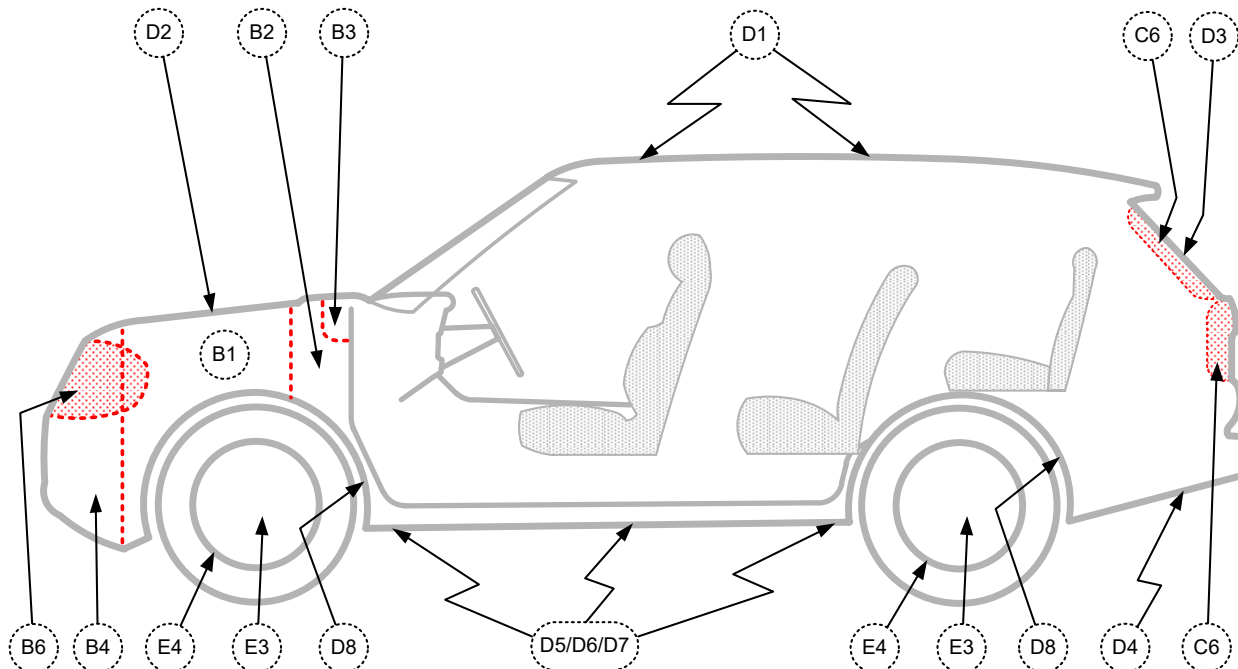


Figure 6 - Exterior Zones for SUV



Title: Electrical and Electronic Component Environmental Requirements

Figure 7 - Interior Zones for Trucks

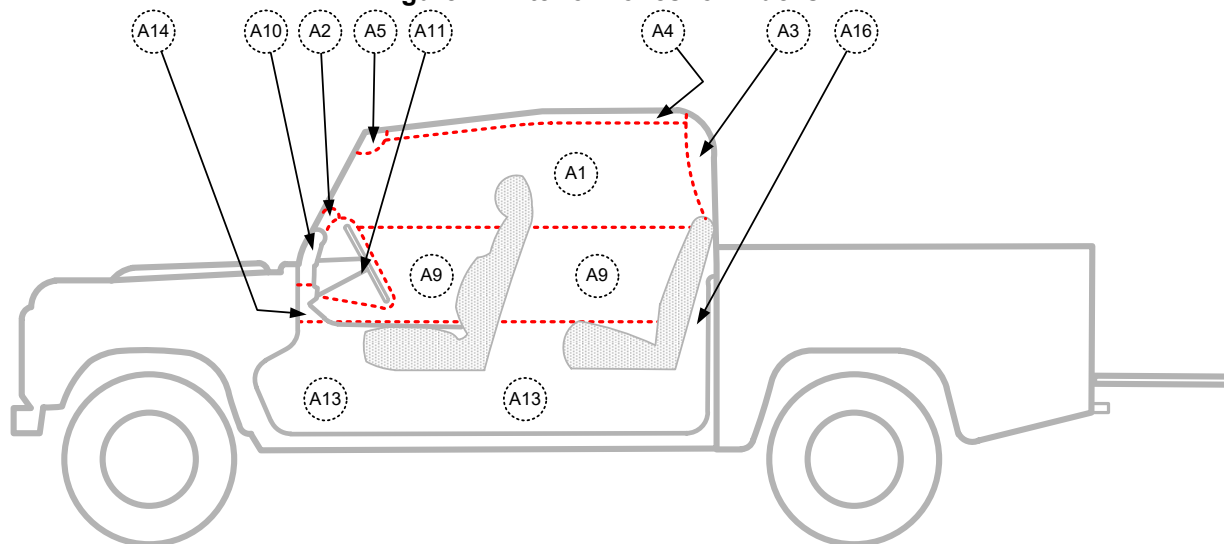
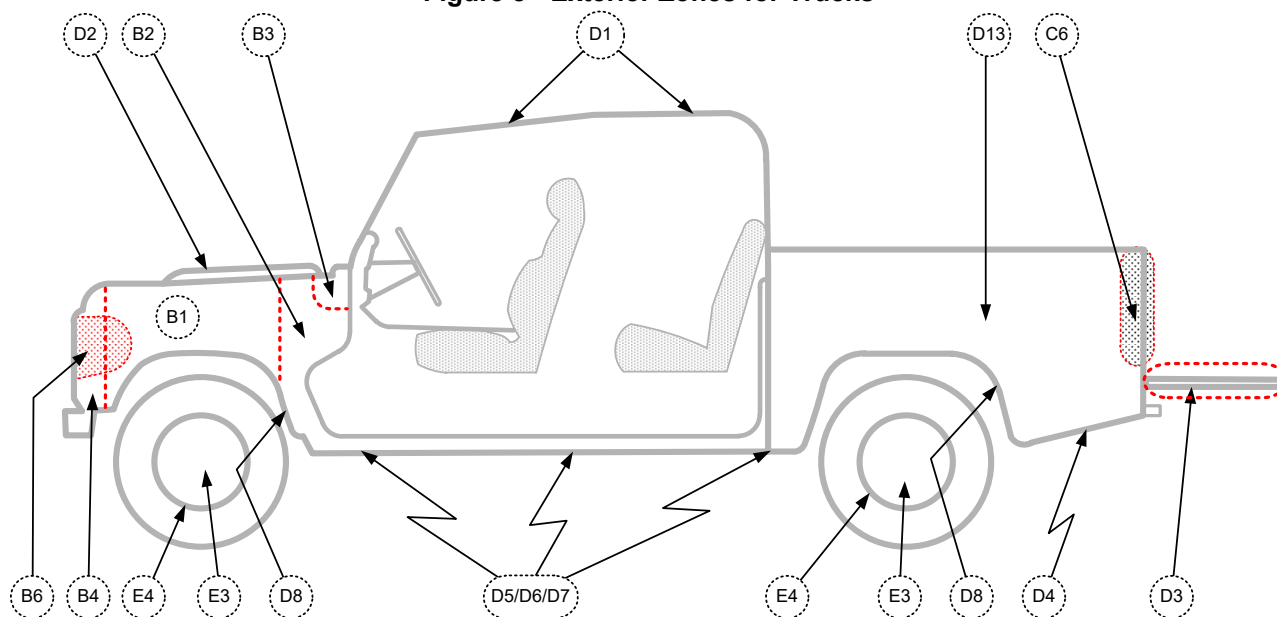
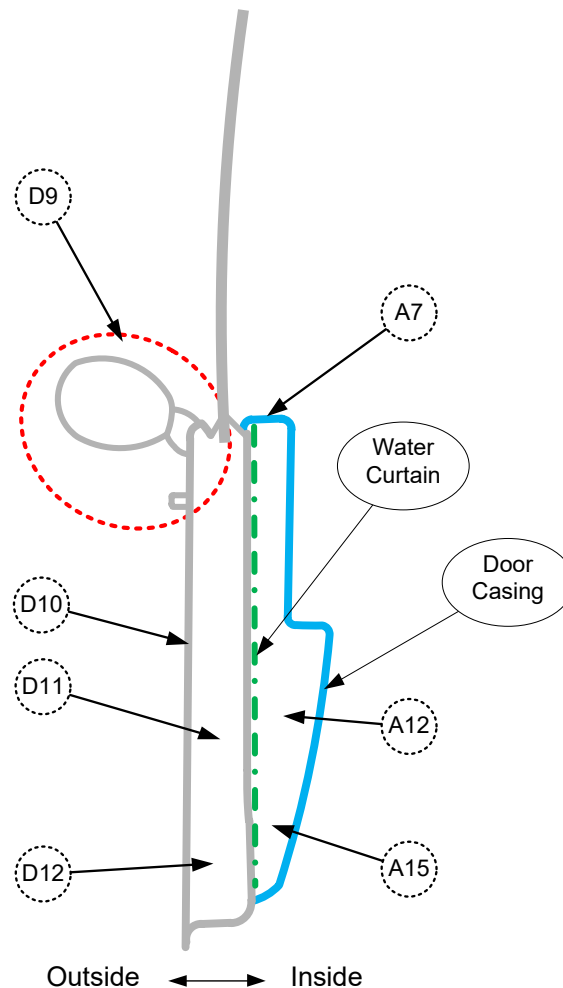


Figure 8 - Exterior Zones for Trucks



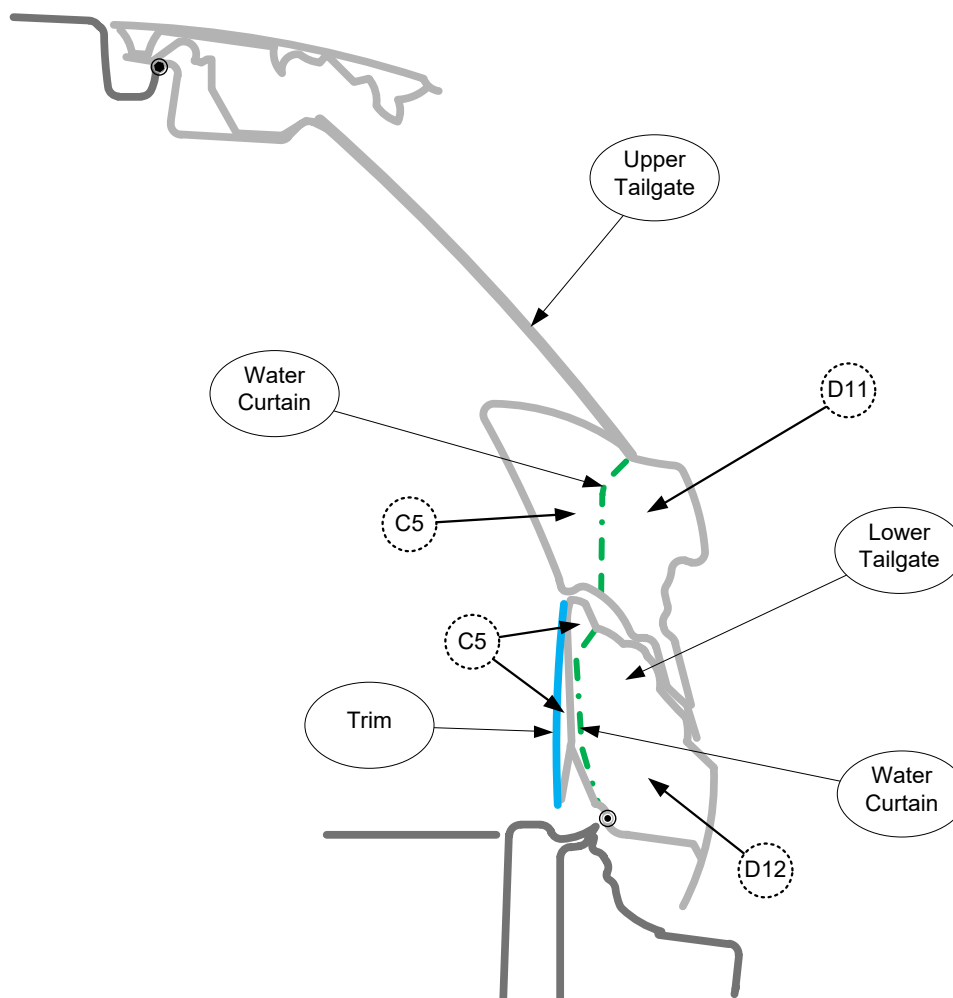
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Figure 9 - Door Frame Zones



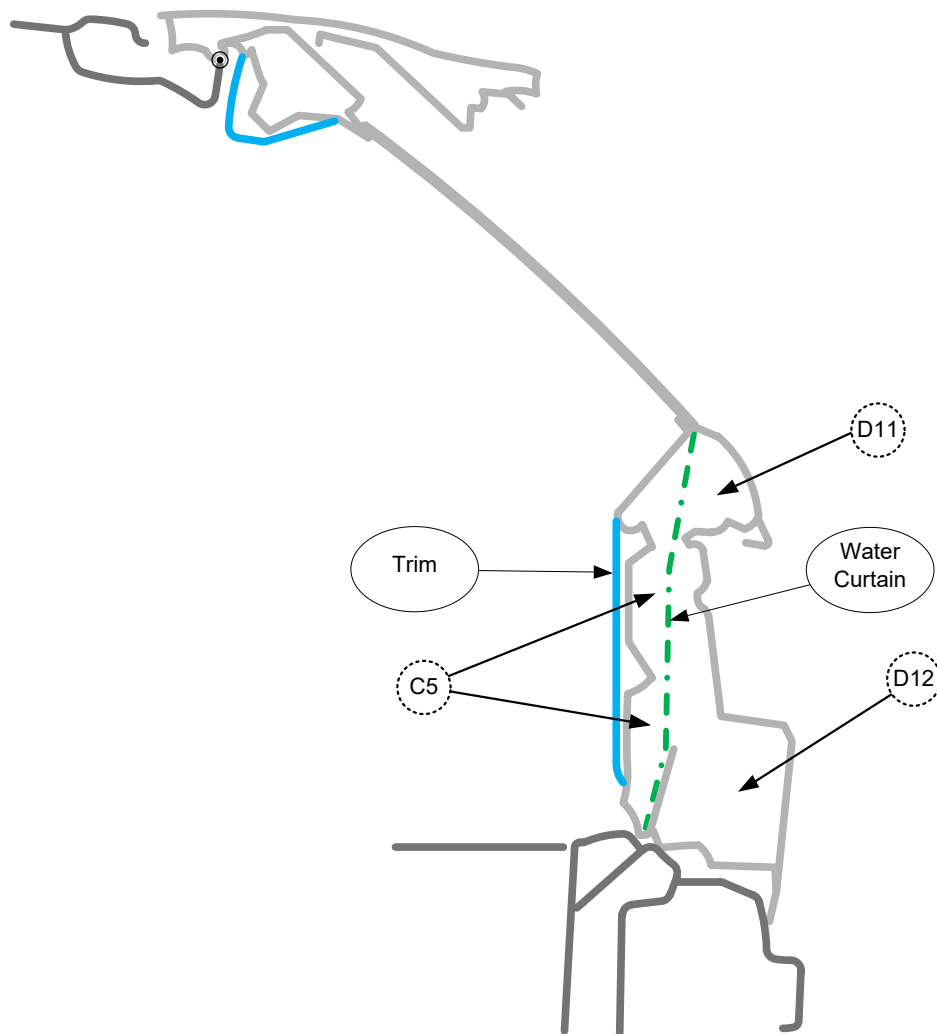
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Figure 10 - Split Tailgate Zones



Title: Electrical and Electronic Component Environmental Requirements

Figure 11 - One Piece Tailgate Zones



Title: Electrical and Electronic Component Environmental Requirements

Figure 12 - Underside of Vehicle Zones

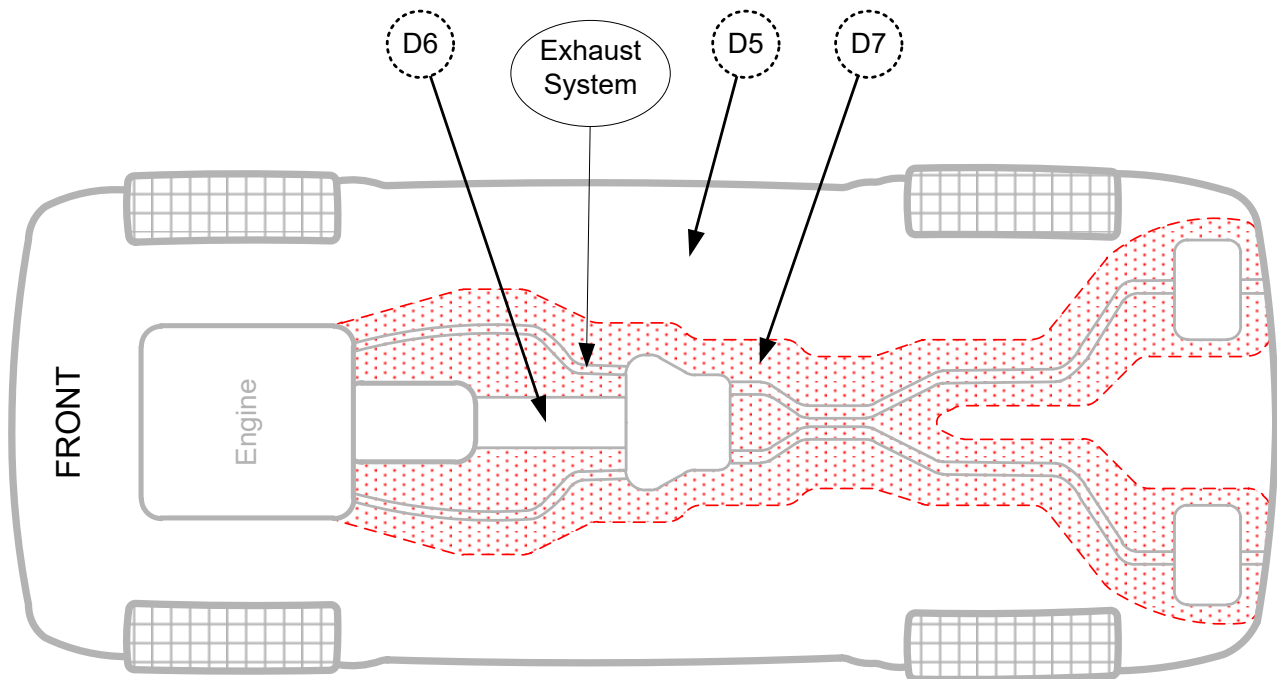
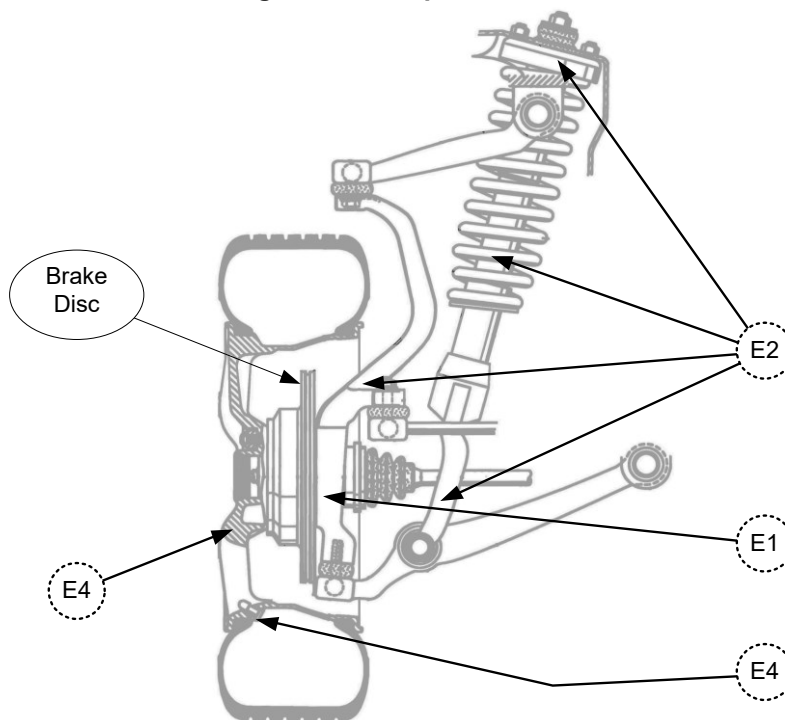


Figure 13 - Suspension and Wheel Zones



Title: Electrical and Electronic Component Environmental Requirements

Table 10 - Summary of Packaging Zones

A	Passenger Compartment
A1	Upper cabin (including head restraints)
A1b	Upper cabin (A/B/C/D Pillars)
A2	Fascia top
A3	Rear parcel shelf
A4	Between Roof and Headlining
A5	Mirror pod (including windscreen mounted components)
A6	Upper tailgate pod
A7	Top of Door Casing
A8	Tailgate centre pod
A9	Mid cabin (between seat cushion and head restraint)
A10	Upper Fascia, Cluster
A11	Steering wheel and column
A12	Within Door Casing (Above Wade-Line/cabin side of Water Curtain)
A13a	Lower Cabin, within the splash zone, as set out in the Interior Wet Zone definition (Section H)
A13b	Lower Cabin, Within the immersion zone, as set out in the Interior Wet Zone Definition (Section H)
A13c	Lower Cabin, not in splash or immersion zones, as set out in the Interior Wet Zone Definition (Section H)
A14	Lower Fascia
A15	Within Door Casing (Below Wade-Line /cabin side of Water Curtain)
A16	Behind rear seats (between seat back and rear bulkhead)
B	Engine Bay
B1	Front, Sides of Engine Bay, Bonnet underside
B2	Rear (behind engine)
B3	Plenum
B4	Forward of cooling pack
B5	On engine / transmission
B6	Headlamp zone (extends between B4 and B1)
C	Load Space
C1	Underside of Boot Lid / Upper Tailgate (when opened & closed)
C2	Between Rear Parcel shelf and trim
C3	Areas outside the splash zone, as set out in the Interior Wet Zone Definition (Section H)
C4a	Areas within the splash zone, as set out in the Interior Wet Zone Definition (Section H)
C4b	Areas within the immersion zone, as set out in the Interior Wet Zone Definition (Section H)
C5	Lower Tailgate (for vehicles with split tailgate) on dry side of water curtain
C6	Rear lamp zone
D	Vehicle Exterior
D1	On roof exterior
D2	On bonnet exterior
D3	On boot / tailgate exterior
D4	Under body - rear (under boot floor, inner rear bumper)
D5	Under Floor - sides
D6	Under Floor - tunnel
D7	Under Floor - within 150 mm of exhaust
D8	Between Wheel-Arch Liners and Body
D9	Door Exterior (handles, mirrors)
D10	Door/Wing Exterior upright panel
D11	Within Door Casing (Above Wade-Line/outside of Water Curtain)
D12	Within Door Casing (Below Wade-Line/outside of Water Curtain)
D13	In Load space area (including truck drop gate)
E	Wheels and Suspension
E1	On hub / Near brake disc
E2	On moving suspension component
E3	On the wheel
E4	On wheel rim/inside the Tyre



Title: Electrical and Electronic Component Environmental Requirements

Z **Change History**

Issue	Date	Description of Change
1	10/02/2021	First issue
2	05/03/2021	Corrections to Test Groups
3	21/01/2022	• Multiple amendments and additions to lifetime, voltage, temperature and vibration requirements. • Amendments to test sequences